

Problem Statement: AI-Driven Financial Distress Prediction for SMEs and Corporate Credit Risk

Financial distress among MSMEs is one of the most pressing credit challenges in India. With 63 million MSMEs contributing approximately 30% of GDP and over ₹5 lakh crore in outstanding MSME loans classified as non-performing assets, the cost of late or inaccurate credit assessment is enormous. Wholesale trading firms — particularly those operating in commodity-intensive sectors such as spices, textiles, and agri-inputs — face unique working capital pressures driven by input price volatility, seasonal demand cycles, and extended payment periods of 30–90 days.

Existing credit scoring systems (CIBIL, bureau-based models) are fundamentally reactive: they detect stress after repayment behavior has already deteriorated. This paper proposes a proactive early warning system that identifies MSME cash flow stress 90 days before it escalates into default, using leading operational signals that existing models ignore — GST filing irregularity, commodity price volatility, payment cycle stretch, and seasonal demand patterns.

The primary users are banks, NBFCs, and credit analysts making three decisions: (1) whether to extend working capital credit to a wholesale trader, (2) how to tier risk across a portfolio of SME borrowers, and (3) what specific operational factors are driving a firm's risk, using SHAP-based explainability for regulatory compliance.

Success is defined as achieving a C-index above 0.72 on survival models, AUC above 0.80 on classification, and precision in the top risk decile above 60% — validated against real historical stress scenarios in India's commodity wholesale sector. The system must be interpretable, deployable via API, and actionable for non-technical credit officers.